

5 Chapter 5: Magnetostatics

5.0 Preliminaries

In the previous chapter, we explored the relationship between electric field, $\vec{E}$, and electric flux density, $\vec{D}$, which are mediated via the property of permittivity, ϵ . For magnetic fields, the subject of this chapter, there is analogous relationship described by

$$\vec{B} = \mu \vec{H},$$

where $\vec{B}$ is the magnetic flux density, $\vec{H}$ is the magnetic field intensity, and μ is the relevant medium's **magnetic permeability**. For materials with constant magnetic permeability μ , the above relationship holds. The difference in magnetic fields versus electric fields is that magnetic fields are only generated by *moving* charges; in other words, current.

5.1 Magnetic Forces and Torques

Charges that *move* in a magnetic field experience a force. Given a point charge, for example, with charge q with velocity $\vec{u}$ in a magnetic flux density $\vec{B}$, then

$$\vec{F}_m = q\vec{u} \times \vec{B} \text{ N},$$

where $\vec{F}_m$ represents the **magnetic force**.

Clearly, if the charge is *stationary*, it will experience no magnetic force, since $\vec{u} = 0 \frac{\text{m}}{\text{s}}$. For example: To find $\vec{F}_m$ on a $+q$ moving at a velocity $\vec{u} = \hat{\alpha}_y u_0$ in a magnetic flux density $\vec{B} = \hat{\alpha}_z B_0$.

$$\begin{aligned}\vec{F}_m &= q\vec{u} \times \vec{B} \\ \vec{F}_m &= q(\hat{\alpha}_y u_0 \times \hat{\alpha}_z B_0) \\ \vec{F}_m &= \hat{\alpha}_x q u_0 B_0 \text{ N}.\end{aligned}$$

Additionally, we must consider the possibility of both magnetic and electric forces on the point charge. The total force on the point charge is simply the sum of the two forces, as described by the Lorents Force equation,

$$\vec{F} = \vec{F}_e + \vec{F}_m = q\vec{E} + q\vec{u} \times \vec{B} = q(\vec{E} + \vec{u} \times \vec{B}) \text{ N}.$$

Since moving point charges experience a force in a magnetic field, we can reasonably assume that current carrying conductors also experience magnetic forces under the same circumstances. Generalizing,

$$\begin{aligned}q\vec{u} &\rightarrow I d\vec{l} = \vec{I} dl \\ d\vec{F}_m &= \vec{I} \times \vec{B} dl \\ \vec{F}_m &= \int d\vec{F}_m = \int_L I d\vec{l} \times \vec{B} = I \int_L d\vec{l} \times \vec{B} = \int_L (\vec{I} \times \vec{B}) dl.\end{aligned}$$

For example: given a semicircular conductor carrying a current I , the closed circuit is exposed to a magnetic field $\vec{B} = \hat{\alpha}_y B_0$, find

- a) The magnetic force $\vec{F}_1$ on the straight section of the wire.

For the straight section (bottom of the semicircle) we can define $d\vec{l} = \hat{\alpha}_x dx$, since the entire section is in the x direction. Then,

$$\begin{aligned}\vec{F}_1 &= I \int d\vec{l} \times \vec{B} = I \int \hat{\alpha}_x dx \times \hat{\alpha}_y B_0 \\ \vec{F}_1 &= I B_0 \hat{\alpha}_z \int_0^{2r} dx = \hat{\alpha}_z 2Ir B_0 \text{ N}.\end{aligned}$$

- b) The magnetic force $\vec{F}_2$ on the curved section. Based on the cross product $d\vec{l} \times \vec{B}$, we know that the force will be in the $-\hat{\alpha}_z$ direction (right-hand rule). Finding $d\vec{l}$, using the cylindrical coordinate system, $d\vec{l} = r d\phi \hat{\alpha}_\phi$. Then,

$$\begin{aligned}\vec{F}_2 &= I \int d\vec{l} \times \vec{B} = I \int r d\phi \hat{\alpha}_\phi \times \hat{\alpha}_y B_0 \\ \vec{F}_2 &= Ir B_0 \int (-\sin(\phi) \hat{\alpha}_x + \cos(\phi) \hat{\alpha}_y) \times \hat{\alpha}_y d\phi \\ \vec{F}_2 &= Ir B_0 (-\hat{\alpha}_z) \int_0^\pi \sin(\phi) d\phi = -\hat{\alpha}_z 2Ir B_0 \text{ N.}\end{aligned}$$

Therefore, $\vec{F}_1 = -\vec{F}_2$, and there is no net force.

Current carrying loops under the influence of magnetic fields can also *rotate* due to magnetic torque. For example, given a magnetic field in the $x - y$ coordinate system $\vec{B} = \hat{\alpha}_y B_y + \hat{\alpha}_z B_z$. Given a current carrying loop (square) in the $x - y$ plane, what current direction will produce torque under the given magnetic field? Testing all the available options,

$$\begin{aligned}I_1 &= \hat{\alpha}_y I \\ I_2 &= -\hat{\alpha}_x I \\ I_3 &= -\hat{\alpha}_y I \\ I_4 &= \hat{\alpha}_x I.\end{aligned}$$

Then, using $\vec{F}_m = \int_L (\vec{I} \times \vec{B}) dl$,

$$\begin{aligned}\vec{F}_1 &= (\hat{\alpha}_y I \times \hat{\alpha}_z B_z) l_1 = \hat{\alpha}_x I B_z l_1 \\ \vec{F}_2 &= (-\hat{\alpha}_x I \times \hat{\alpha}_z B_z) l_2 = \hat{\alpha}_y I B_z l_1 \\ \vec{F}_3 &= (-\hat{\alpha}_y I \times \hat{\alpha}_z B_z) l_1 = -\hat{\alpha}_x I B_z l_1 \\ \vec{F}_4 &= (\hat{\alpha}_x I \times \hat{\alpha}_z B_z) l_2 = -\hat{\alpha}_y I B_z l_2\end{aligned}$$

Therefore, for the normal (B_z) component of the field, the loop is pulled in all directions equally, and no torque results. Finding the effect of the B_y components,

$$\begin{aligned}\vec{F}_1 &= (\hat{\alpha}_y I \times \hat{\alpha}_y B_y) l_1 = 0 \\ \vec{F}_2 &= (-\hat{\alpha}_x I \times \hat{\alpha}_y B_y) l_2 = -\hat{\alpha}_z I B_y l_1 \\ \vec{F}_3 &= (-\hat{\alpha}_y I \times \hat{\alpha}_y B_y) l_1 = 0 \\ \vec{F}_4 &= (\hat{\alpha}_x I \times \hat{\alpha}_y B_y) l_2 = \hat{\alpha}_z I B_y l_2.\end{aligned}$$

Therefore, net force is still zero, but now, since $\vec{F}_1$ and $\vec{F}_2$ are equal but in opposite directions, there will be a torque that causes rotation in *what direction*?

More generally, torque can be found as $\vec{m} = IA \hat{\alpha}_n$, where $\vec{m}$ is the vector magnetic moment, A is the loop area, and $\hat{\alpha}_n$ is the unit vector normal to the loop (right-hand rule applied to current direction). Then,

$$\vec{T} = \vec{m} \times \vec{B},$$

where $\vec{T}$ is the direction of torque. Alternately,

$$\begin{aligned}\vec{T} &= \hat{\alpha}_T m B \sin(\theta) \\ \vec{T} &= \hat{\alpha}_T I A B \sin(\theta).\end{aligned}$$

5.2 The Biot-Savart Law

The Biot-Savart law defines the static magnetic field produced by a steady current.

To formulate the Biot-Savart law, we take a short section $d\vec{l}$ of a current carrying wire, and define the resultant magnetic field at a point lying at a straight-line distance R from the current carrying wire,

$$d\vec{H} = \frac{I}{4\pi} \frac{d\vec{l} \times \hat{\alpha}_r}{R^2}.$$

Integrating over the entire current carrying wire,

$$\vec{H} = \int d\vec{H} = \frac{I}{4\pi} \int \frac{d\vec{l} \times \hat{\alpha}_r}{R^2},$$

where $\hat{\alpha}_r = \frac{\vec{R}}{|\vec{R}|}$.

Alternately,

$$\vec{H} = \frac{I}{4\pi} \int \frac{d\vec{l} \times \vec{R}}{R^3} \frac{\text{A}}{\text{M}}. \text{ (check these units)}$$

The direction of the resultant field will be in the ϕ direction around the wire.